

In [ ]: Name:Srushti Rajendra Mulik RollNo:2010

In [ ]: Seaborn Mastery:The Complete Data Visualization Training Notebook

## 1.Imports & Environment Setup

```
In [3]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

sns.set_theme(style="whitegrid",palette="deep")
plt.rcParams['figure.dpi']=100

print("Libraries imported successfully")
print("Seaborn version:",sns.__version__)
print("pandas version:",pd.__version__)
```

Libraries imported successfully  
Seaborn version: 0.13.2  
pandas version: 2.3.1

## 2.Dataset Loading

```
In [5]: # Load the dataset
df=sns.load_dataset("tips")
# Alternative if you have a local file:
# df=pd.read_csv("tips.csv")

df
```

Out[5]:

|     | total_bill | tip  | sex    | smoker | day  | time   | size |
|-----|------------|------|--------|--------|------|--------|------|
| 0   | 16.99      | 1.01 | Female | No     | Sun  | Dinner | 2    |
| 1   | 10.34      | 1.66 | Male   | No     | Sun  | Dinner | 3    |
| 2   | 21.01      | 3.50 | Male   | No     | Sun  | Dinner | 3    |
| 3   | 23.68      | 3.31 | Male   | No     | Sun  | Dinner | 2    |
| 4   | 24.59      | 3.61 | Female | No     | Sun  | Dinner | 4    |
| ... | ...        | ...  | ...    | ...    | ...  | ...    | ...  |
| 239 | 29.03      | 5.92 | Male   | No     | Sat  | Dinner | 3    |
| 240 | 27.18      | 2.00 | Female | Yes    | Sat  | Dinner | 2    |
| 241 | 22.67      | 2.00 | Male   | Yes    | Sat  | Dinner | 2    |
| 242 | 17.82      | 1.75 | Male   | No     | Sat  | Dinner | 2    |
| 243 | 18.78      | 3.00 | Female | No     | Thur | Dinner | 2    |

244 rows × 7 columns

### 3.Data Understanding

#### 3.1 df.head()-peek at the First Rows

In [6]: `df.head()`

Out[6]:

|   | total_bill | tip  | sex    | smoker | day | time   | size |
|---|------------|------|--------|--------|-----|--------|------|
| 0 | 16.99      | 1.01 | Female | No     | Sun | Dinner | 2    |
| 1 | 10.34      | 1.66 | Male   | No     | Sun | Dinner | 3    |
| 2 | 21.01      | 3.50 | Male   | No     | Sun | Dinner | 3    |
| 3 | 23.68      | 3.31 | Male   | No     | Sun | Dinner | 2    |
| 4 | 24.59      | 3.61 | Female | No     | Sun | Dinner | 4    |

In [ ]: 3.2 `df.info()`-Structural Summary

In [7]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   total_bill  244 non-null    float64
1   tip         244 non-null    float64
2   sex         244 non-null    category
3   smoker      244 non-null    category
4   day         244 non-null    category
5   time        244 non-null    category
6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

#### 3.3 df.shape-Dimensions

In [8]: `df.shape`

Out[8]: (244, 7)

#### 3.4 df.columns-Column Names

In [9]: `df.columns`

Out[9]: Index(['total\_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')

#### 3.5 df.dtypes-data types only

In [10]: `df.dtypes`

```
Out[10]: total_bill    float64
         tip          float64
         sex          category
         smoker       category
         day          category
         time         category
         size         int64
         dtype: object
```

### 3.6 df.describe()-Statistical Summary(Numeric Columns)

```
In [11]: df.describe()
```

```
Out[11]:
```

|              | total_bill | tip        | size       |
|--------------|------------|------------|------------|
| <b>count</b> | 244.000000 | 244.000000 | 244.000000 |
| <b>mean</b>  | 19.785943  | 2.998279   | 2.569672   |
| <b>std</b>   | 8.902412   | 1.383638   | 0.951100   |
| <b>min</b>   | 3.070000   | 1.000000   | 1.000000   |
| <b>25%</b>   | 13.347500  | 2.000000   | 2.000000   |
| <b>50%</b>   | 17.795000  | 2.900000   | 2.000000   |
| <b>75%</b>   | 24.127500  | 3.562500   | 3.000000   |
| <b>max</b>   | 50.810000  | 10.000000  | 6.000000   |

### 3.7 df.isnull().sum()-Missing Value Check

```
In [12]: df.isnull().sum()
```

```
Out[12]: total_bill    0
         tip          0
         sex          0
         smoker       0
         day          0
         time         0
         size         0
         dtype: int64
```

### 3.8 df.duplicated().sum()-Duplicate Row Check

```
In [13]: df.duplicated().sum()
```

```
Out[13]: np.int64(1)
```

### 3.9.value\_counts()-Frequency of Categories

```
In [14]: df['day'].value_counts()
```

```
Out[14]: day
Sat      87
Sun      76
Thur     62
Fri      19
Name: count, dtype: int64
```

```
In [ ]: 3.10 .unique()-List Distinct Values
```

```
In [15]: df['day'].unique()
```

```
Out[15]: ['Sun', 'Sat', 'Thur', 'Fri']
Categories (4, object): ['Thur', 'Fri', 'Sat', 'Sun']
```

```
In [ ]: 3.11 .nunique()-Count of Distinct Values
```

```
In [16]: for col in df.columns:
          print(f"{col:12s}->{df[col].nunique()}unique values")
```

```
total_bill ->229unique values
tip        ->123unique values
sex        ->2unique values
smoker     ->2unique values
day        ->4unique values
time       ->2unique values
size       ->6unique values
```

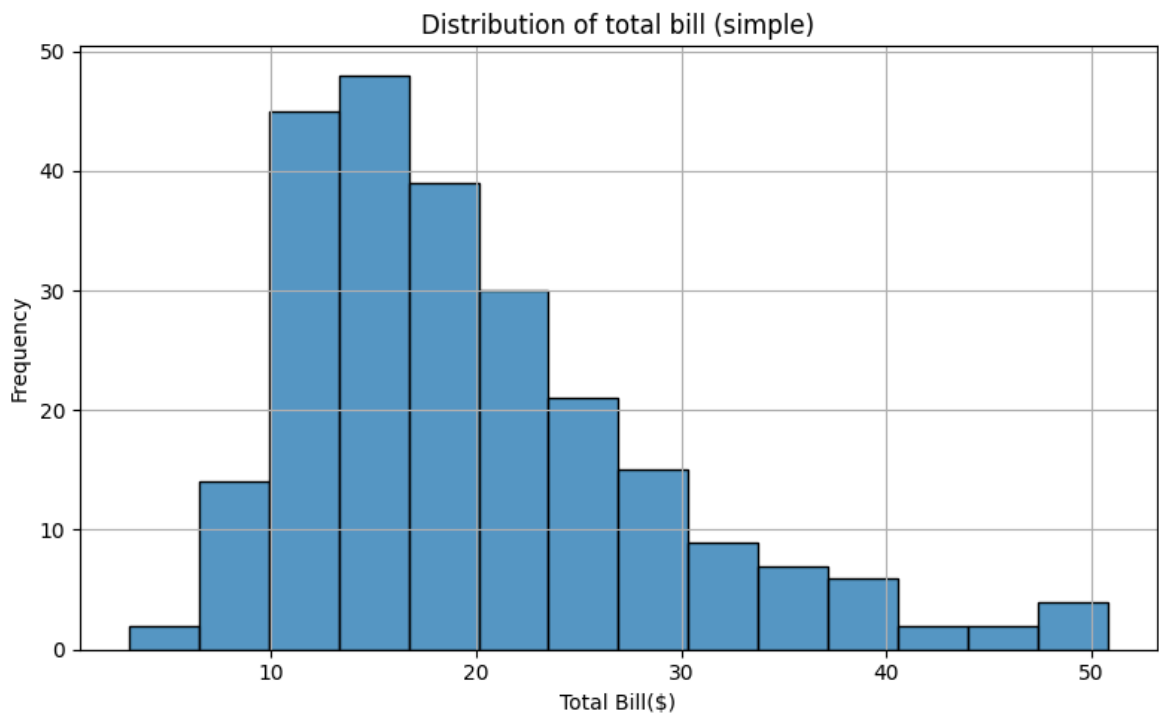
5.Histogram-sns.histplot()/sns.displot

```
In [2]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [4]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [5]: plt.figure(figsize=(8,5))
sns.histplot(df['total_bill'])
plt.title('Distribution of total bill (simple)')
plt.xlabel('Total Bill($)')
plt.ylabel('Frequency')
plt.grid(True)
plt.tight_layout()
plt.show()
```



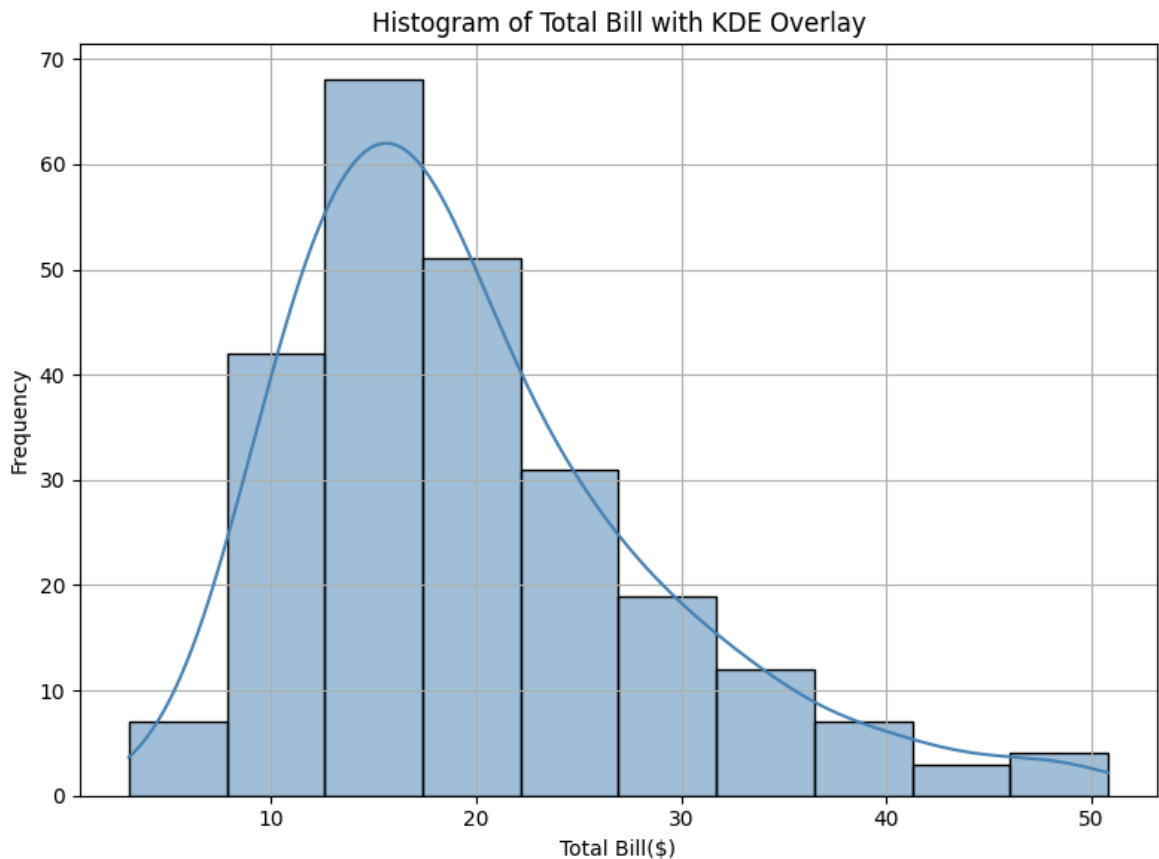
8.intermediate example-with bins&KDE

```
In [4]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [4]: plt.figure(figsize=(8,6))
sns.histplot(df['total_bill'],bins=10,kde=True,color='steelblue')
plt.title('Histogram of Total Bill with KDE Overlay')
plt.xlabel('Total Bill($)')
plt.ylabel('Frequency')
plt.grid(True)
plt.tight_layout()
plt.show()
```



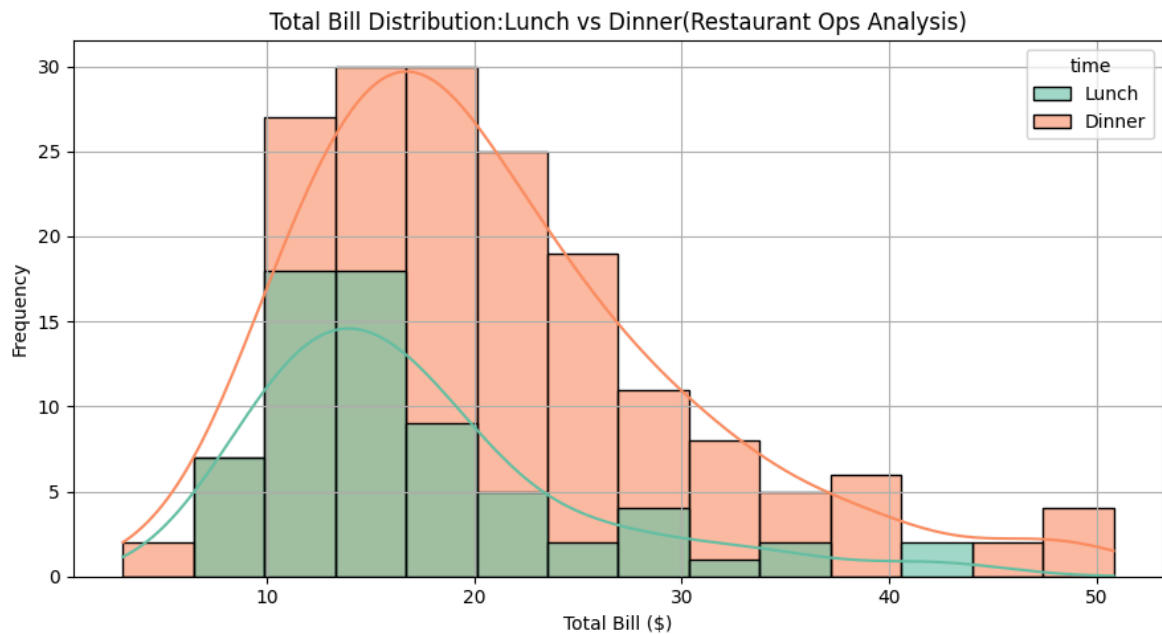
In [ ]: 9.Real industry Example

```
In [6]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [7]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [9]: plt.figure(figsize=(9,5))
sns.histplot(data=df,x='total_bill',hue='time',kde=True,multiple='layer',palette
plt.title('Total Bill Distribution:Lunch vs Dinner(Restaurant Ops Analysis)')
plt.xlabel('Total Bill ($)')
plt.ylabel('Frequency')
plt.grid(True)
plt.tight_layout()
plt.show()
```



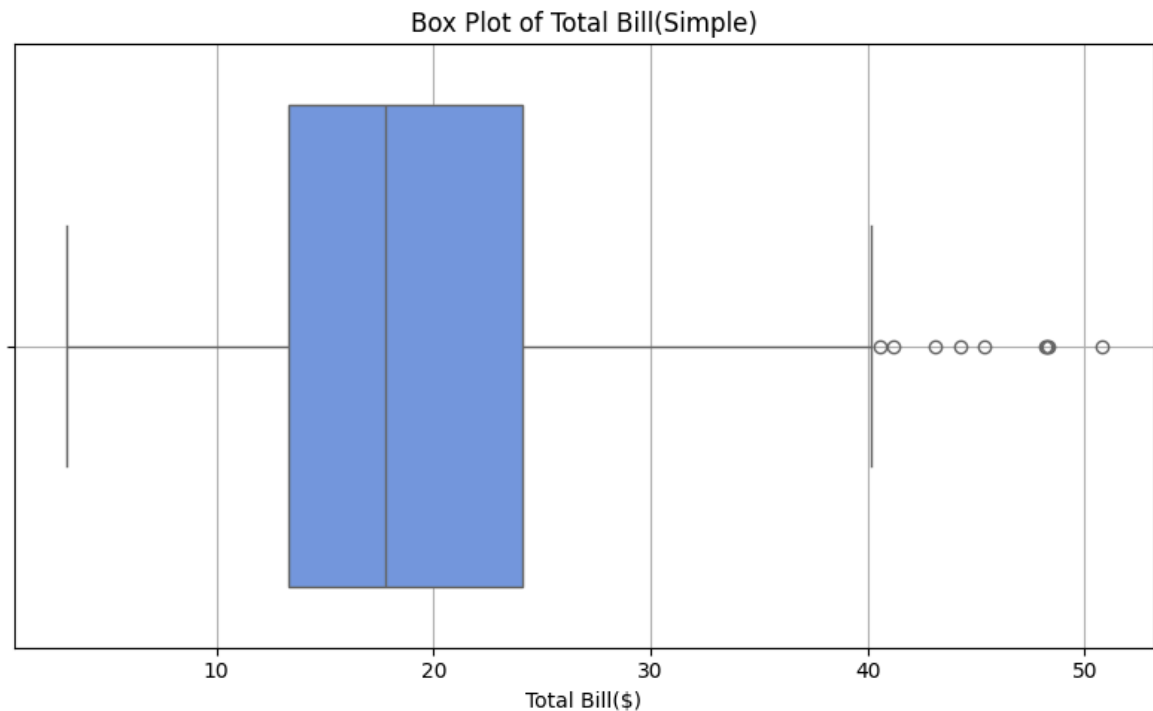
6.Box Plot-sns.boxplot()

```
In [7]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [8]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [6]: plt.figure(figsize=(8,5))
sns.boxplot(x=df['total_bill'],color='cornflowerblue')
plt.title('Box Plot of Total Bill(Simple)')
plt.xlabel('Total Bill($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```



### 9.Real Industry Example-Banking

```
In [6]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [7]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

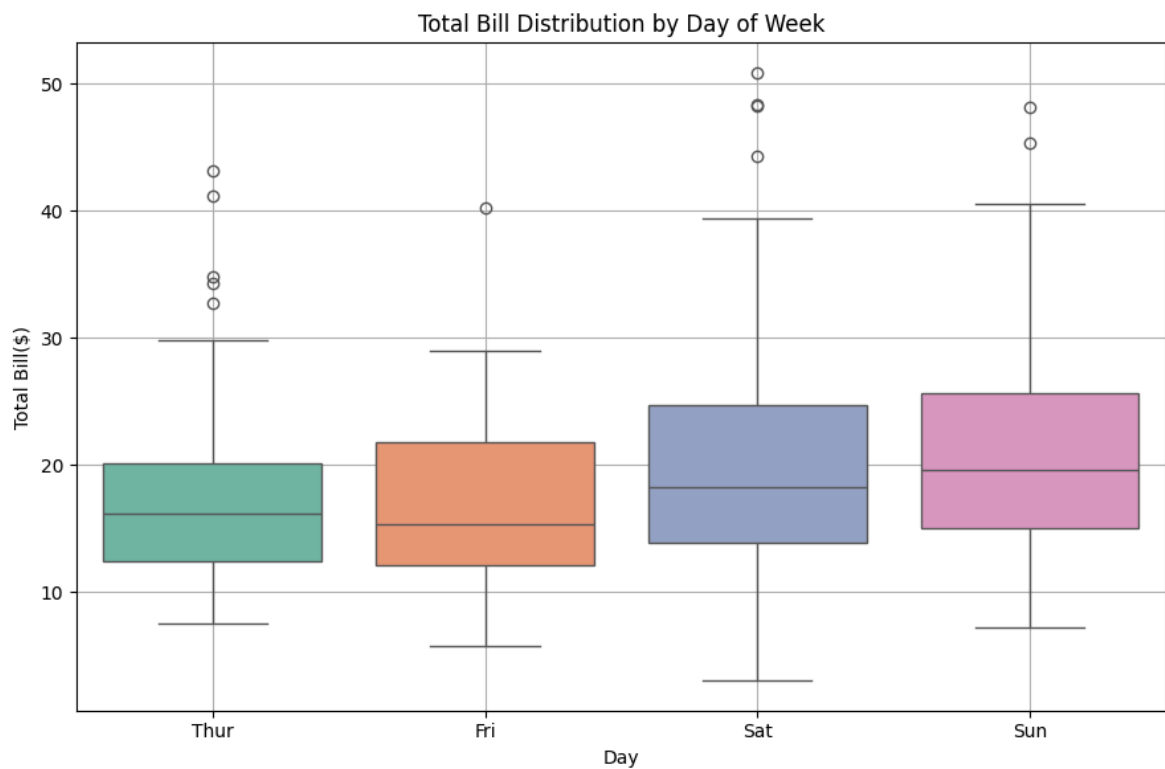
```
In [9]: plt.figure(figsize=(9,6))
sns.boxplot(data=df,x='day',y='total_bill',palette='Set2')
plt.title('Total Bill Distribution by Day of Week')
plt.xlabel('Day')
plt.ylabel('Total Bill($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```

C:\Users\Admin\AppData\Local\Temp\ipykernel\_2056\3000181577.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df,x='day',y='total_bill',palette='Set2')
```





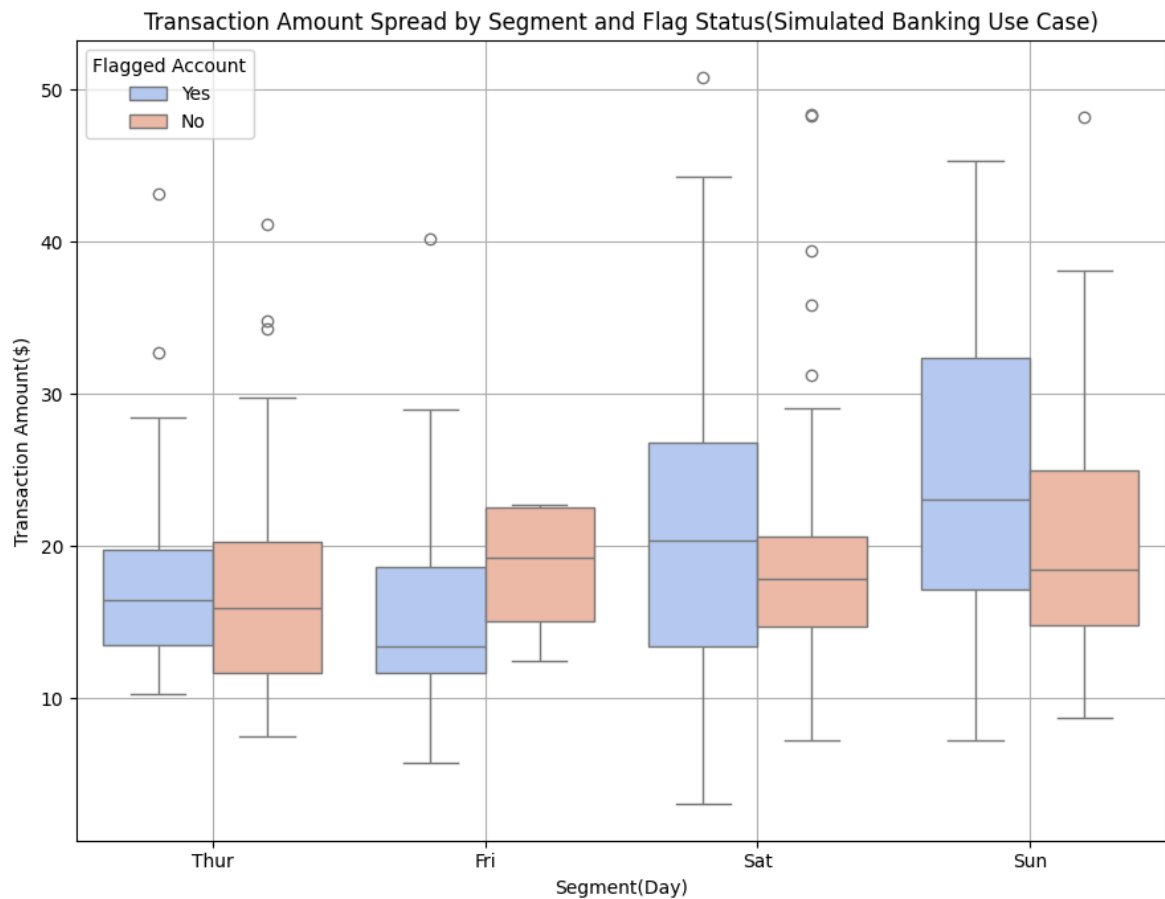
In [ ]: 9.Real Industry Example-Banking

```
In [10]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [11]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [14]: plt.figure(figsize=(9,7))
sns.boxplot(data=df,x='day',y='total_bill',hue='smoker',palette='coolwarm',dodge
plt.title('Transaction Amount Spread by Segment and Flag Status(Simulated Bankin
plt.xlabel('Segment(Day)')
plt.ylabel('Transaction Amount($)')
plt.legend(title='Flagged Account')
plt.grid(True)
plt.tight_layout()
plt.show()
```



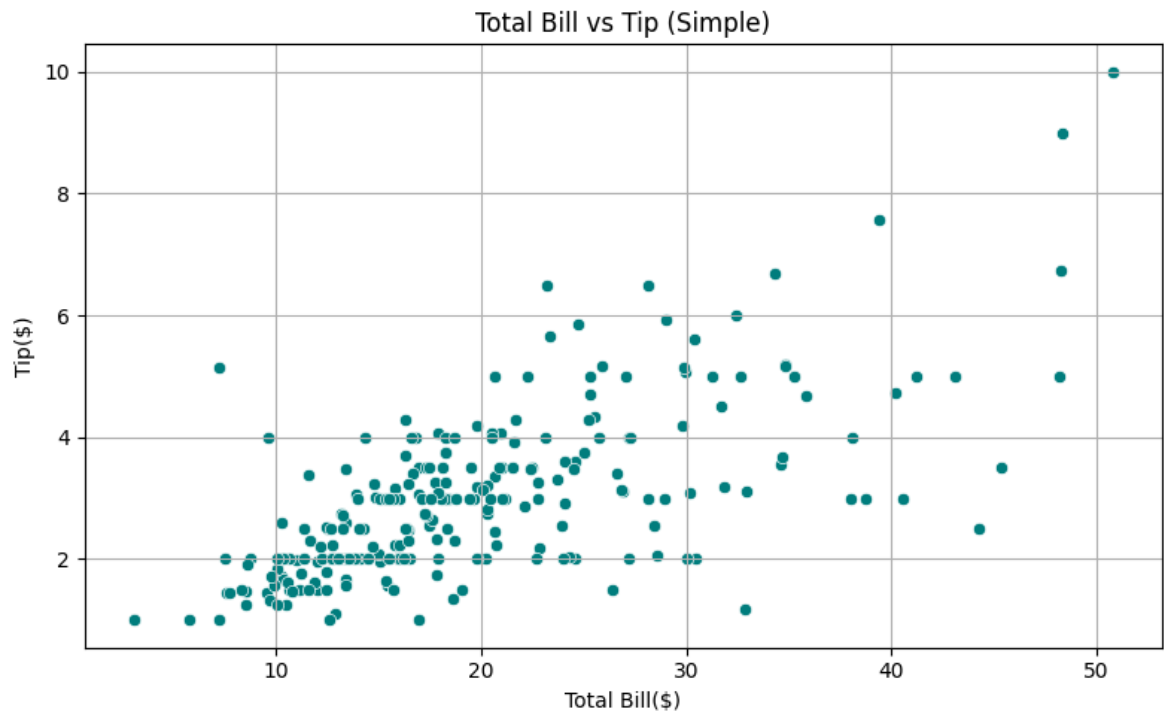
### 7.Scatter Plot-sns.scatterplot()

```
In [15]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [16]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [17]: plt.figure(figsize=(8,5))
sns.scatterplot(x=df['total_bill'],y=df['tip'],color='teal')
plt.title('Total Bill vs Tip (Simple)')
plt.xlabel('Total Bill($)')
plt.ylabel('Tip($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```

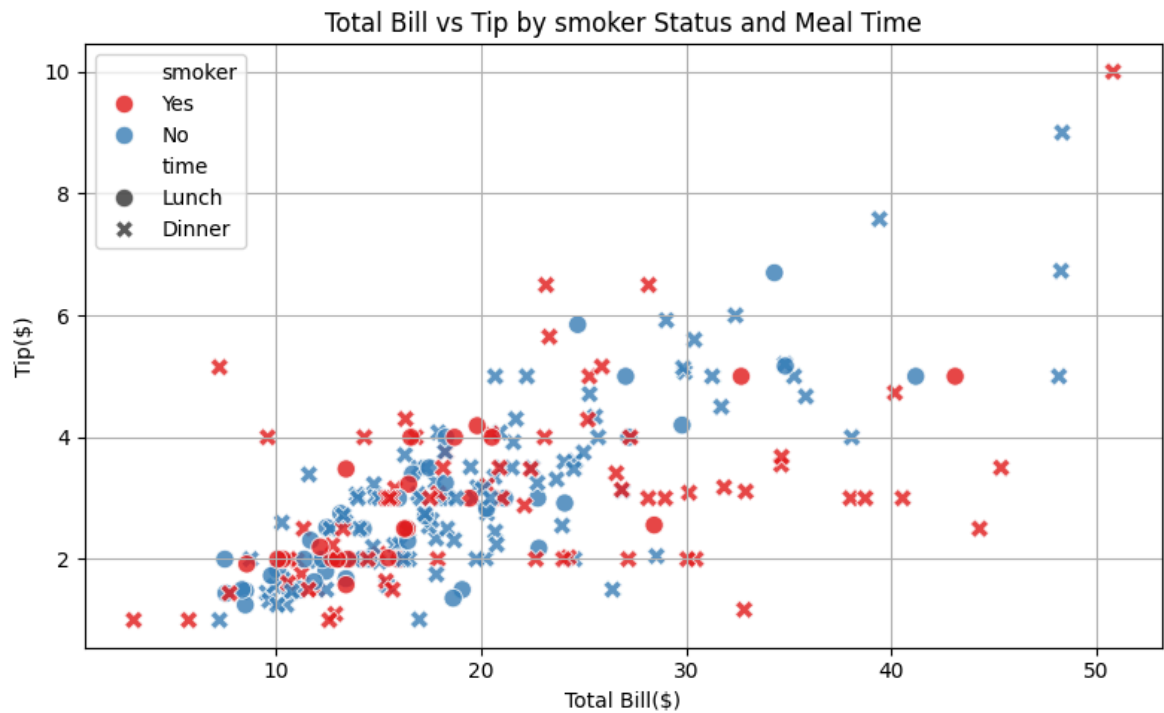


```
In [18]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [19]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [21]: plt.figure(figsize=(8,5))
sns.scatterplot(data=df,x='total_bill',y='tip',hue='smoker',style='time',palette
plt.title('Total Bill vs Tip by smoker Status and Meal Time')
plt.xlabel('Total Bill($)')
plt.ylabel('Tip($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```



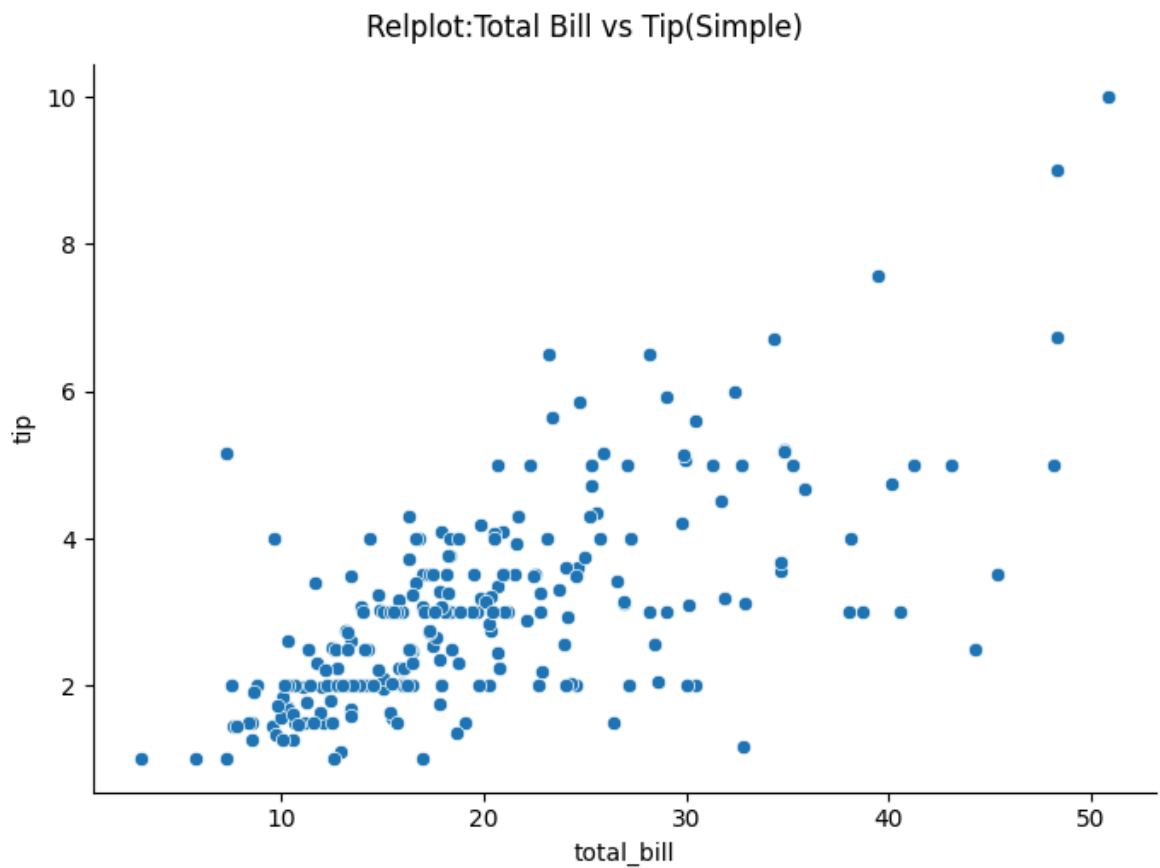
8.Relplot-sns.relplot()

```
In [2]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [6]: sns.relplot(data=df,x='total_bill',y='tip',height=5,aspect=1.4)
plt.suptitle('Relplot:Total Bill vs Tip(Simple)',y=1.03)
plt.show()
```



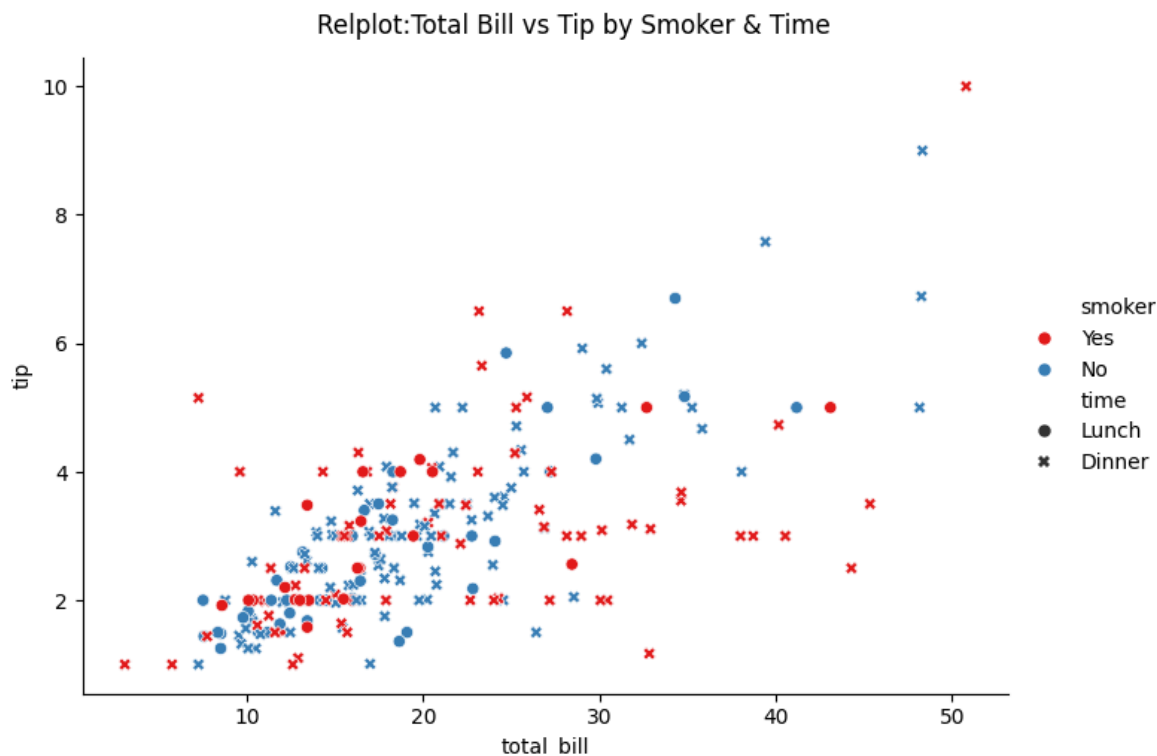
```
In [ ]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [ ]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [7]: sns.relplot(data=df,x='total_bill',y='tip',hue='smoker',style='time',palette='Se

plt.suptitle('Relplot:Total Bill vs Tip by Smoker & Time',y=1.03)
plt.show()
```

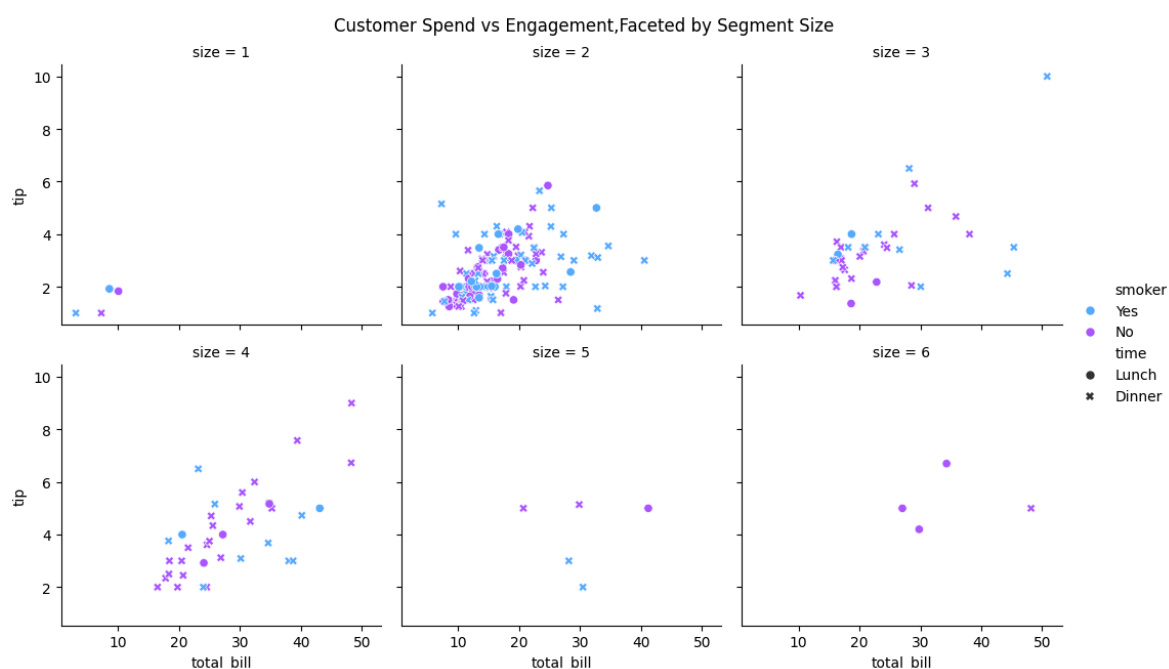


```
In [18]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [19]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [20]: sns.relplot(data=df,x='total_bill',y='tip',hue='smoker',style='time',col='size',
plt.suptitle('Customer Spend vs Engagement,Faceted by Segment Size',y=1.02)
plt.show())
```



## 9.Line Plot-sns.lineplot

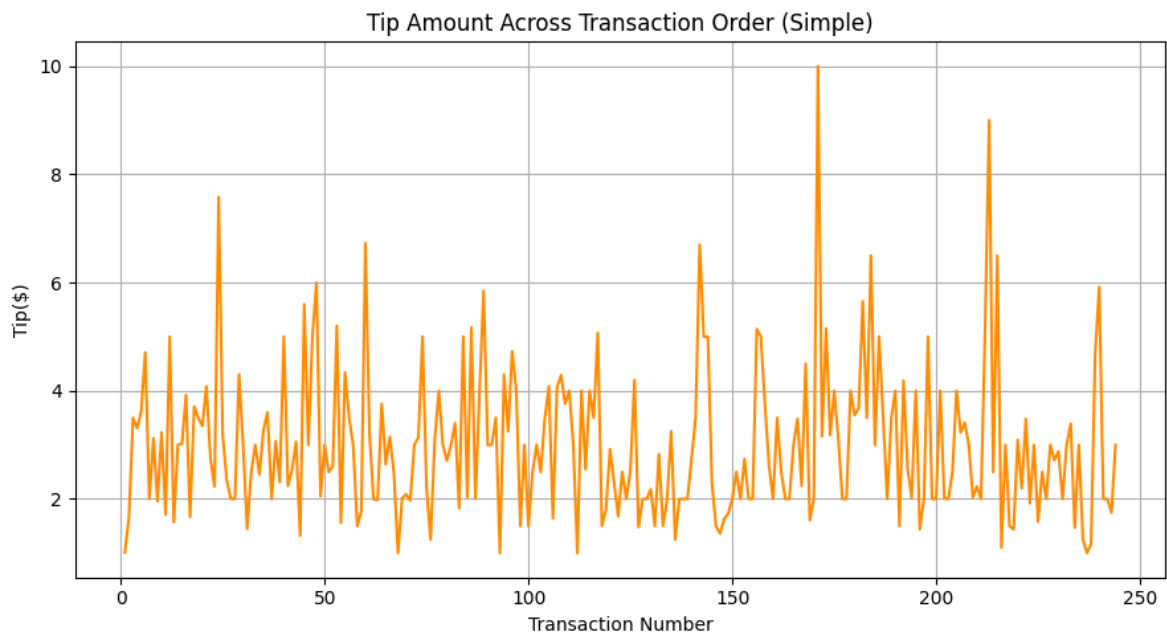
```
In [21]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [24]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [25]: df_sorted=df.copy()
df_sorted['sno']=np.arange(1,len(df_sorted)+1)

plt.figure(figsize=(9,5))
sns.lineplot(x='sno',y='tip',data=df_sorted,color='darkorange')
plt.title('Tip Amount Across Transaction Order (Simple)')
plt.xlabel('Transaction Number')
plt.ylabel('Tip($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```

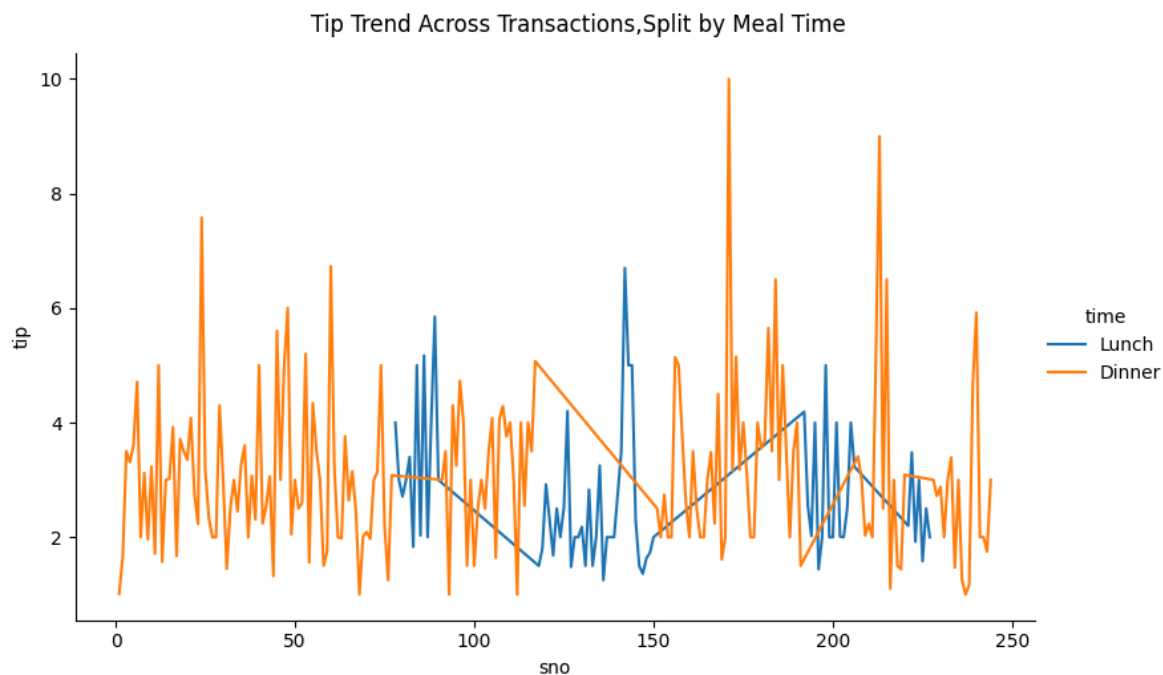


```
In [26]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [27]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [28]: sns.relplot(data=df_sorted,x='sno',y='tip',kind='line',hue='time',height=5,aspect
plt.suptitle('Tip Trend Across Transactions,Split by Meal Time',y=1.03)
plt.show()
```

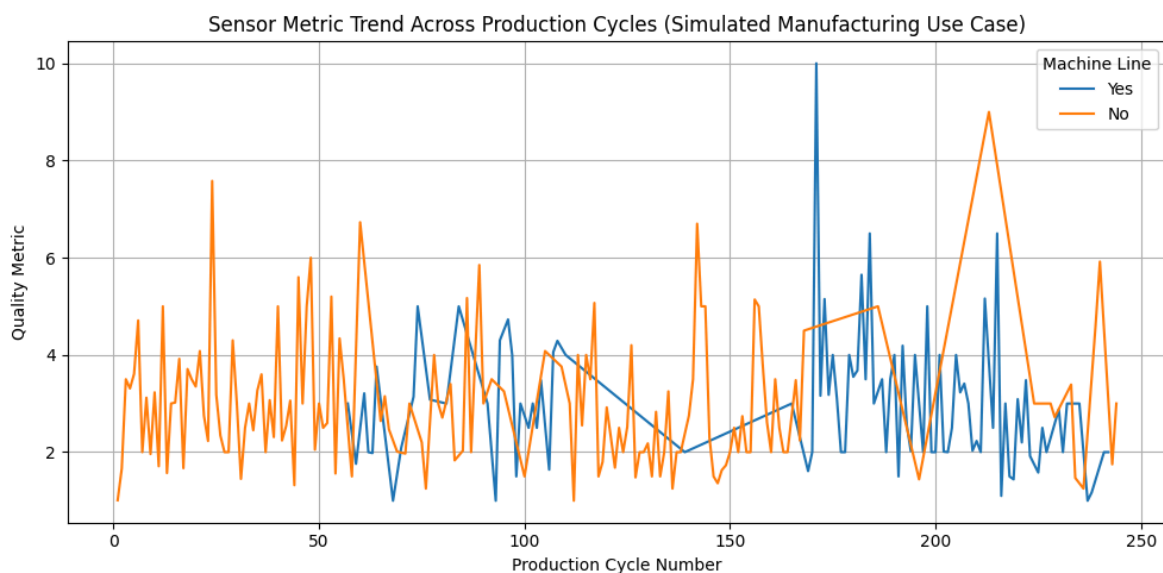


```
In [29]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [30]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [31]: plt.figure(figsize=(10,5))
sns.lineplot(x='sno',y='tip',data=df_sorted,hue='smoker',estimator='mean',errorb
plt.title('Sensor Metric Trend Across Production Cycles (Simulated Manufacturing
plt.xlabel('Production Cycle Number')
plt.ylabel('Quality Metric')
plt.legend(title='Machine Line')
plt.grid(True)
plt.tight_layout()
plt.show()
```





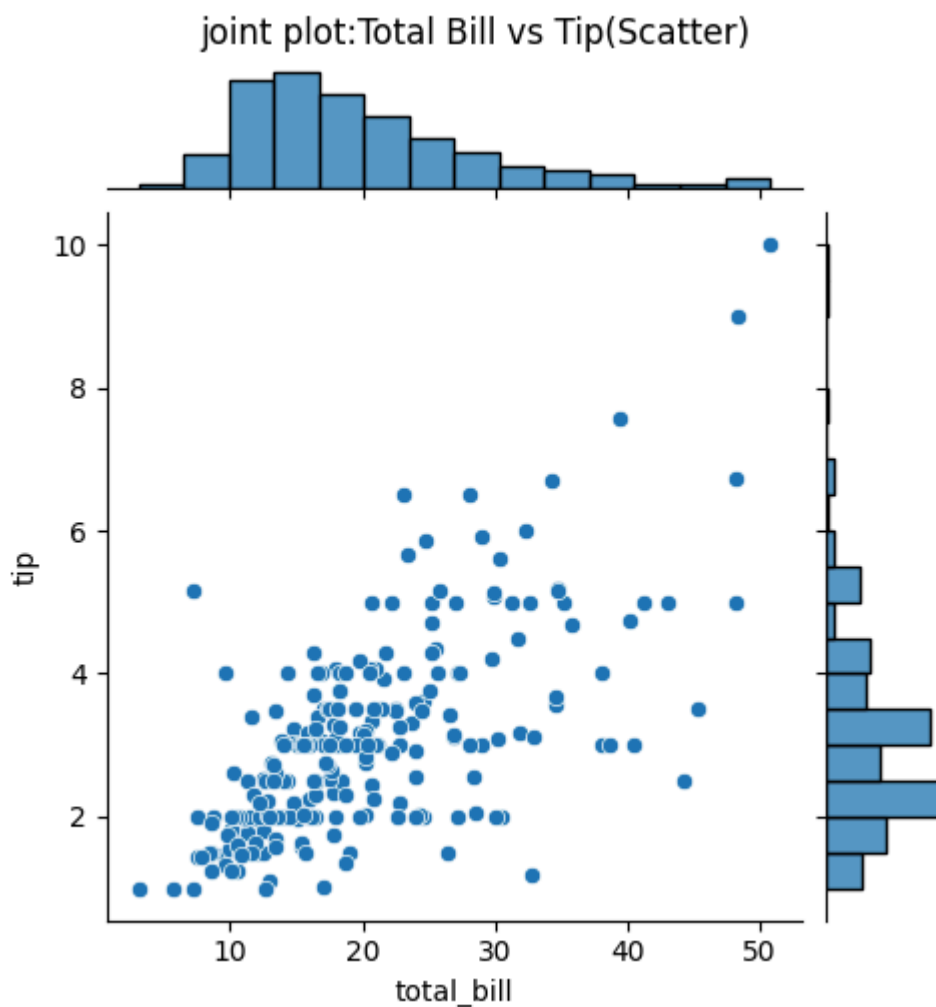
## 10.Joint Plot-sns.jointplot()

```
In [32]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [33]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [34]: sns.jointplot(data=df,x='total_bill',y='tip',height=5)
plt.suptitle('joint plot:Total Bill vs Tip(Scatter)',y=1.02)
plt.show()
```



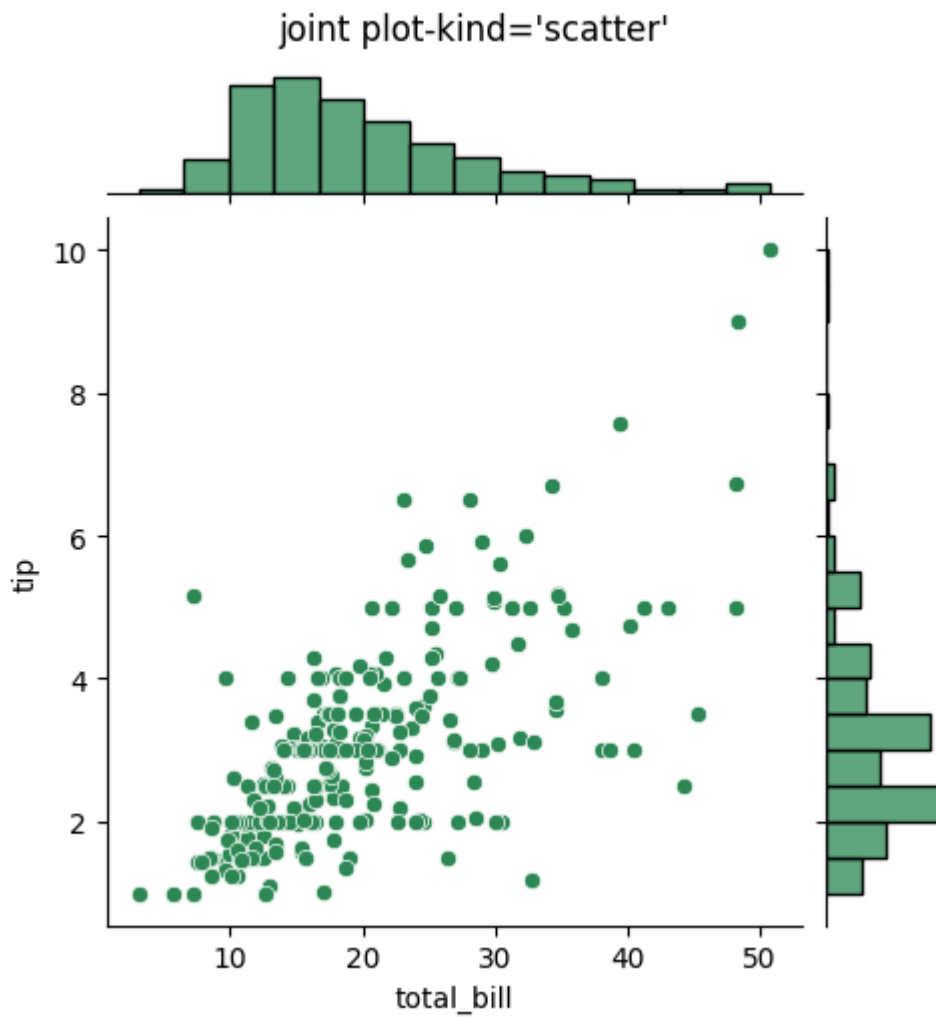
```
In [35]: import matplotlib.pyplot as plt
import seaborn as sns
```

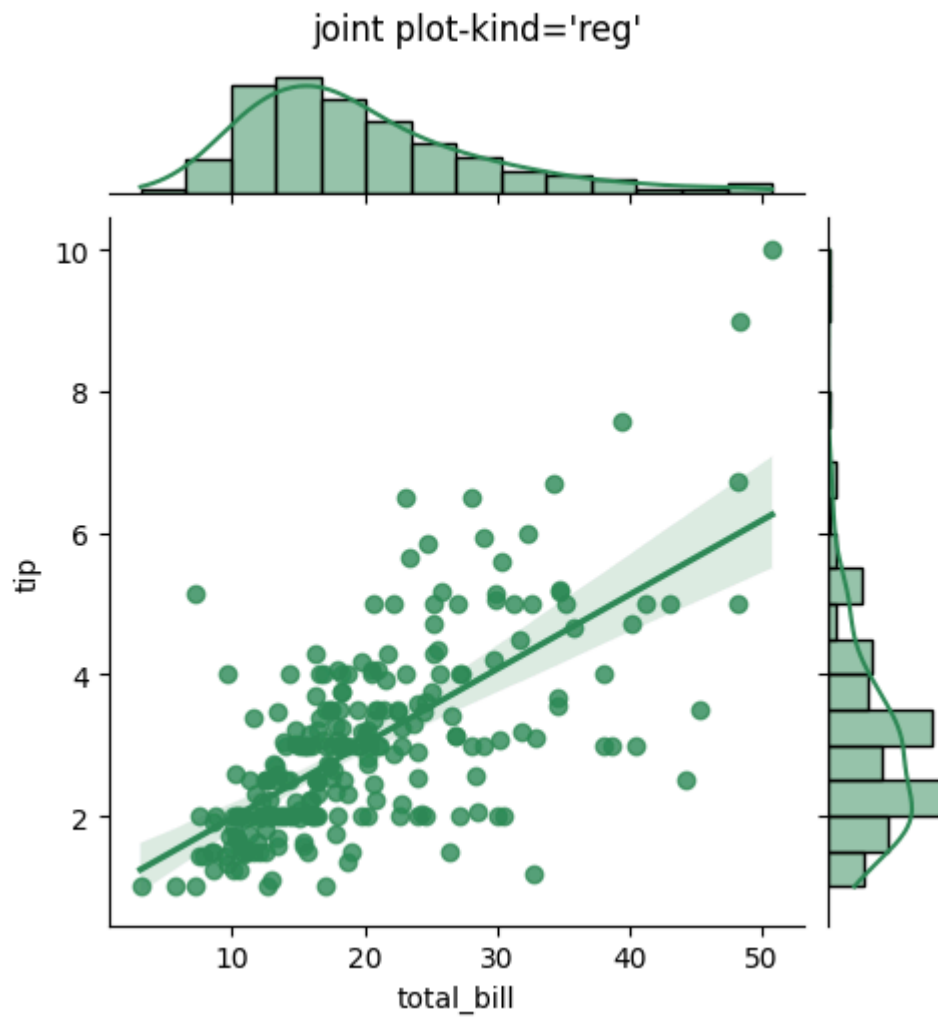
```
In [36]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

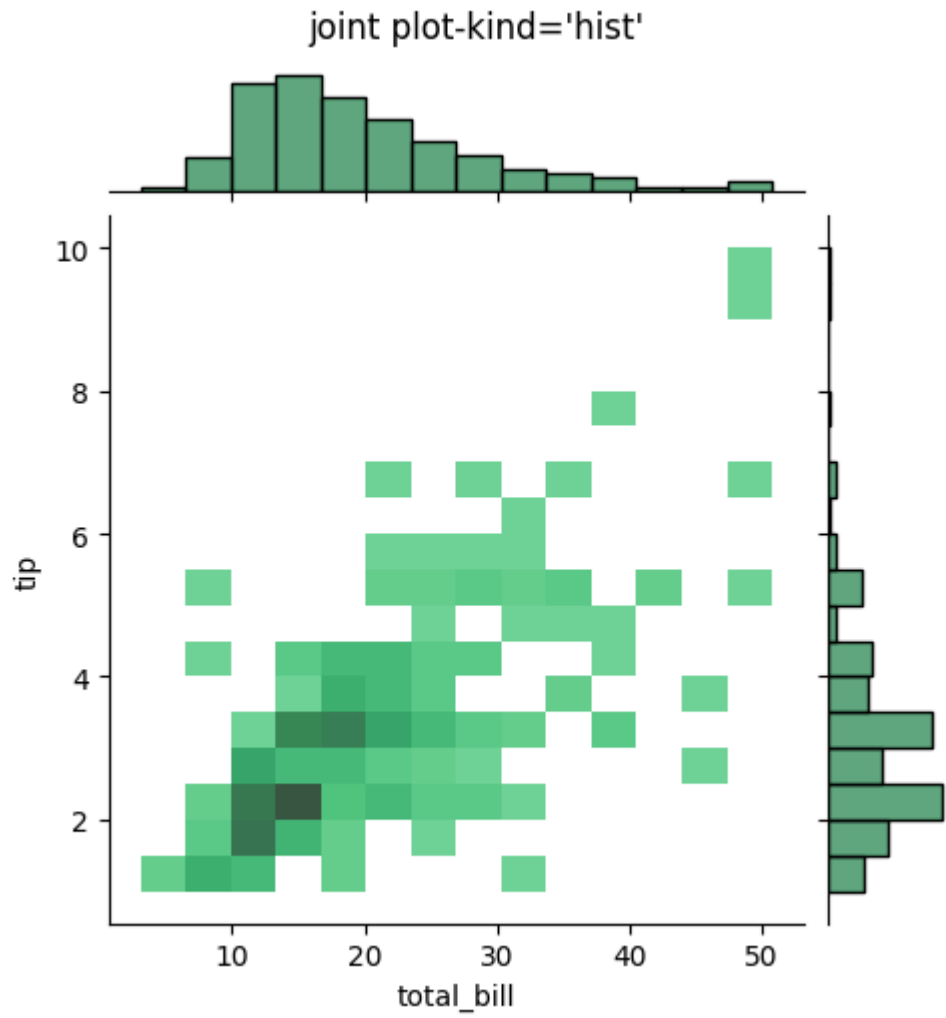
df=sns.load_dataset("tips")
```

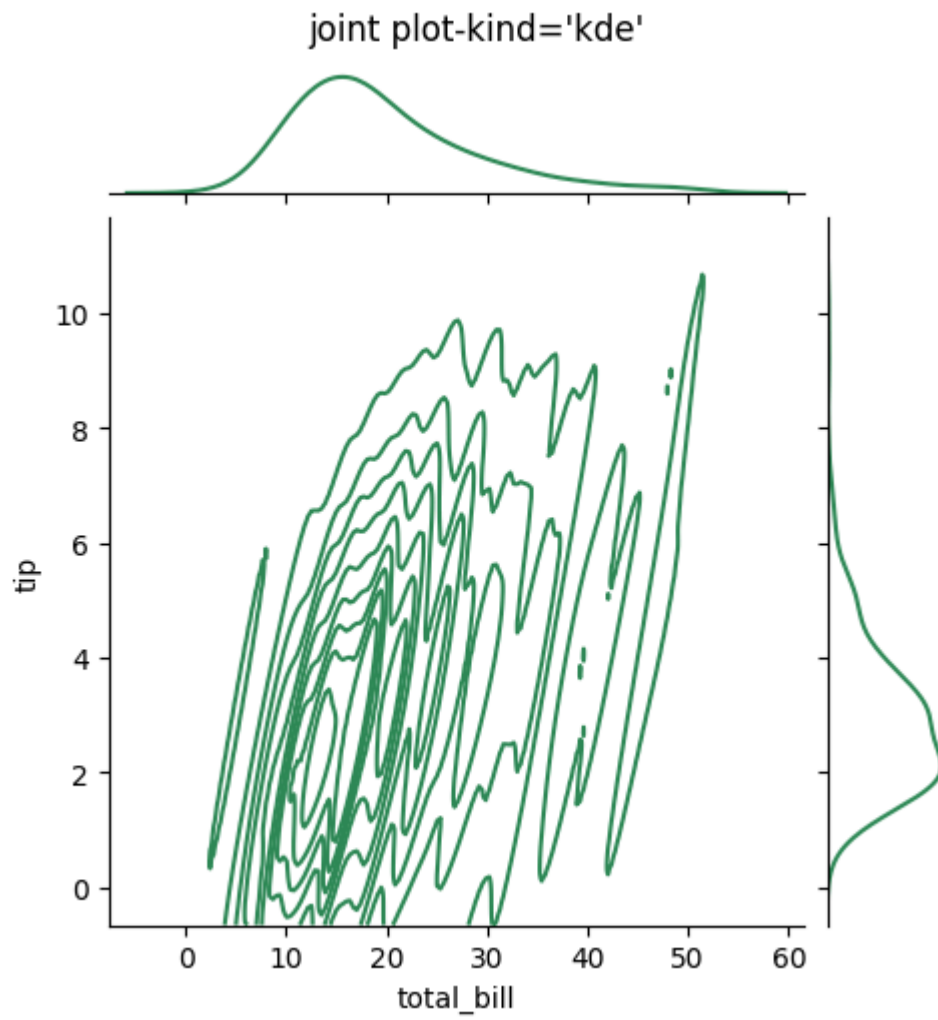
```
In [37]: fig_kinds=['scatter','reg','hist','kde','hex']

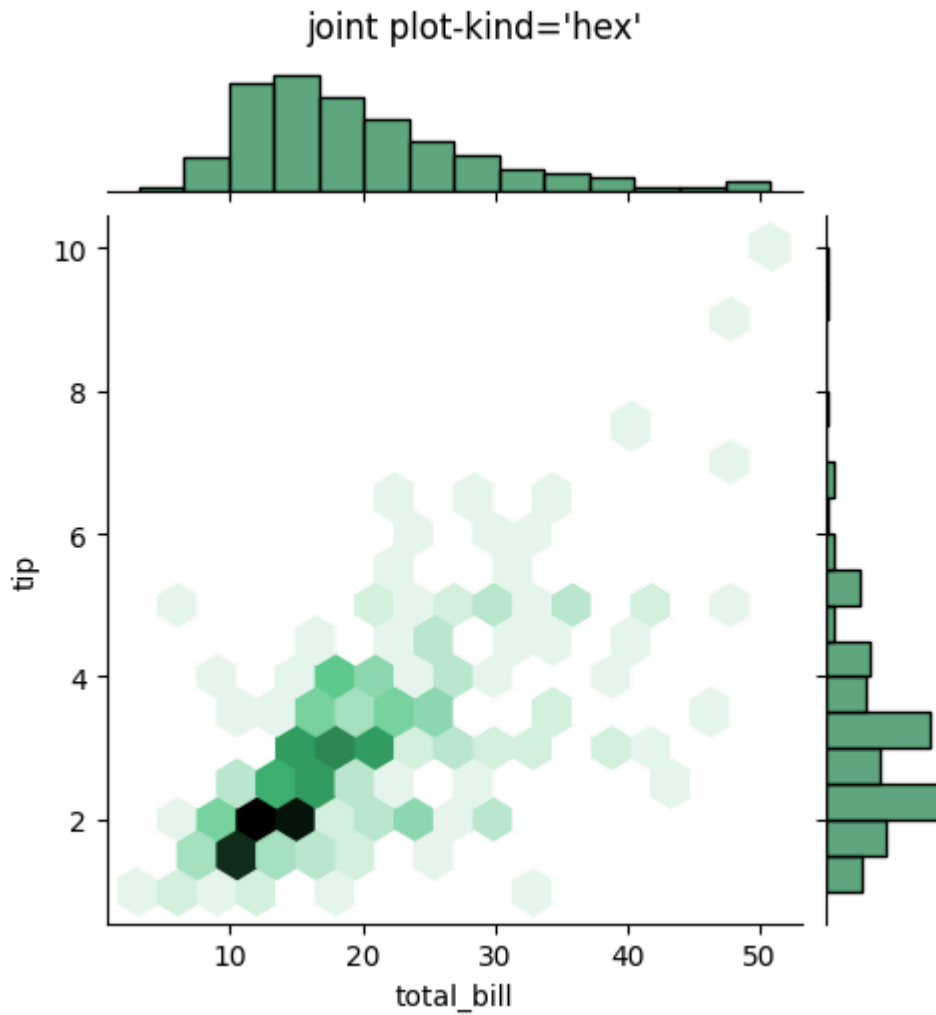
for k in fig_kinds:
    g=sns.jointplot(data=df,x='total_bill',y='tip',kind=k,height=5,color='seagreen')
    g.fig.suptitle(f"joint plot-kind='{k}'",y=1.03)
    plt.show()
```











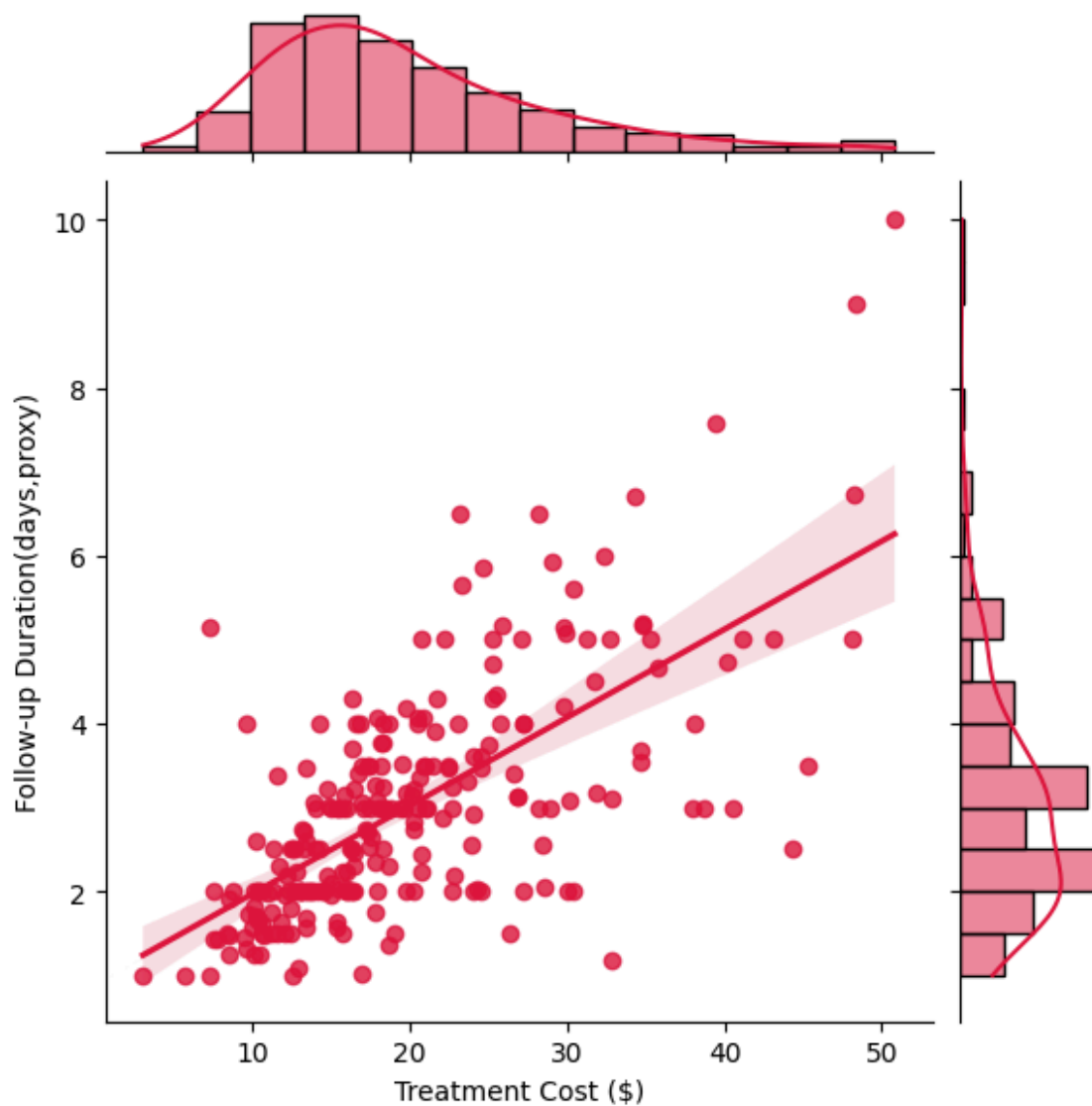
```
In [38]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [39]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [40]: g=sns.jointplot(data=df,x='total_bill',y='tip',kind='reg',height=6,color='crimson')
g.fig.suptitle('Treatment Cost vs Follow-up Duration(Simulated Healthcare Use Ca
g.ax_joint.set_xlabel('Treatment Cost ($)')
g.ax_joint.set_ylabel('Follow-up Duration(days,proxy)')
plt.show()
```

## Treatment Cost vs Follow-up Duration(Simulated Healthcare Use Case)



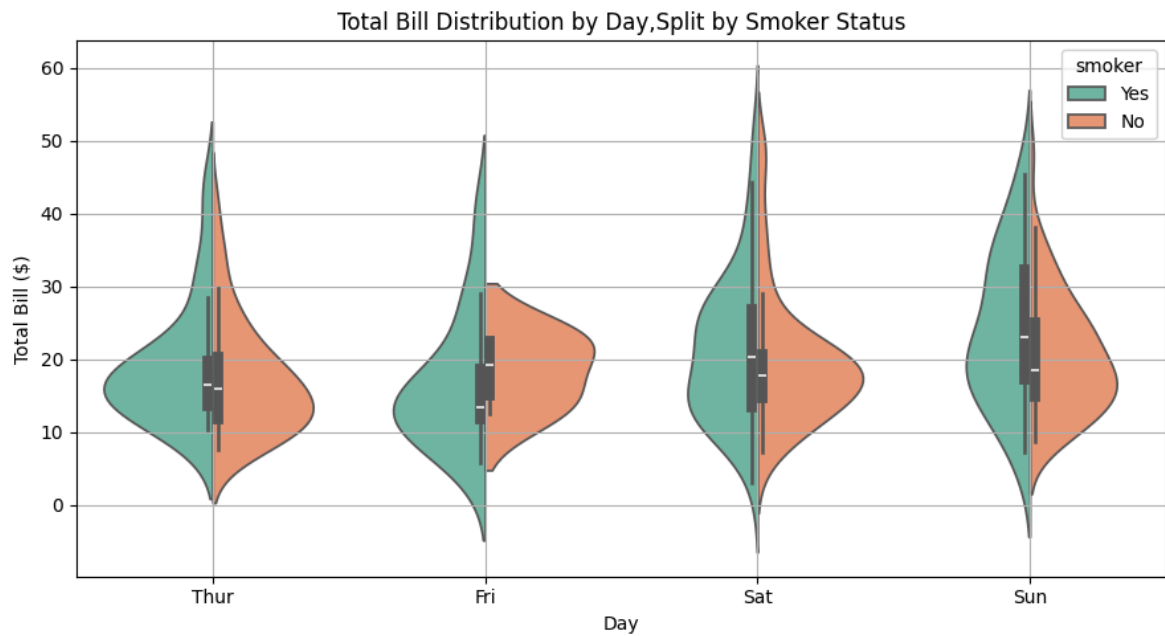
11.Violin plot-sns.violinplot()

```
In [41]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [42]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [44]: plt.figure(figsize=(9,5))
sns.violinplot(data=df,x='day',y='total_bill',hue='smoker',split=True,palette='S
plt.title('Total Bill Distribution by Day,Split by Smoker Status')
plt.xlabel('Day')
plt.ylabel('Total Bill ($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```



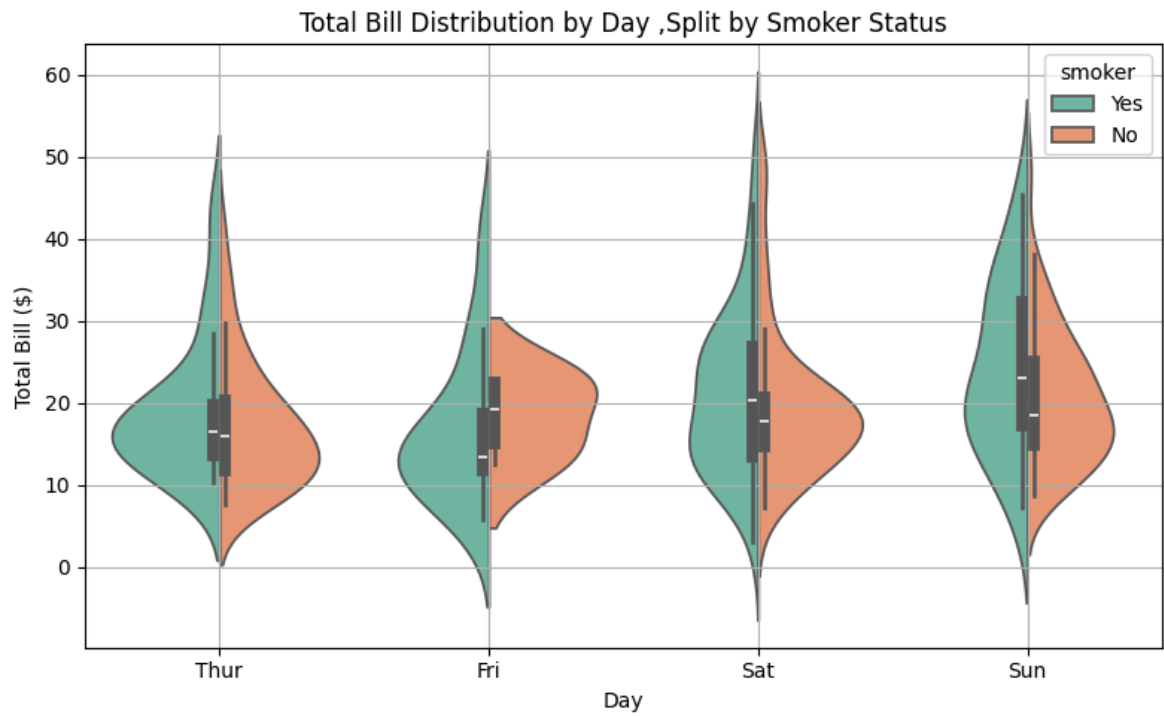
```
In [45]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [46]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [47]: plt.figure(figsize=(8,5))
sns.violinplot(data=df,x='day',y='total_bill',hue='smoker',split=True,palette='S
plt.title('Total Bill Distribution by Day ,Split by Smoker Status')
plt.xlabel('Day')
plt.ylabel('Total Bill ($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```



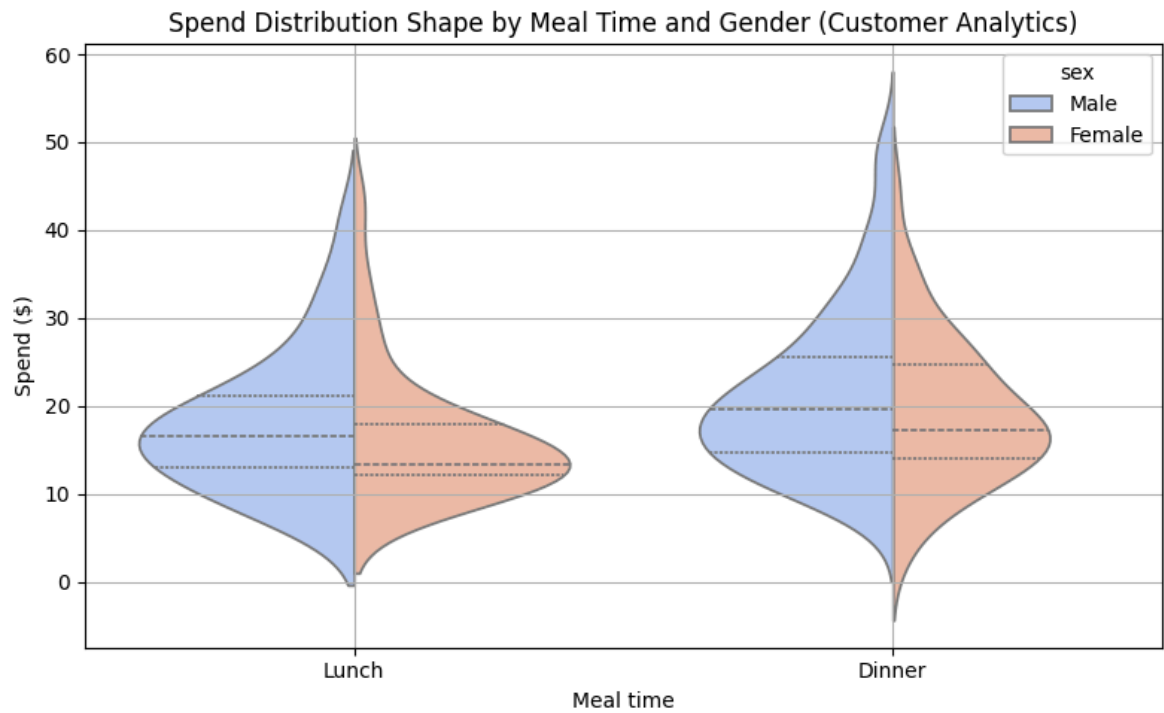


```
In [48]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [49]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [50]: plt.figure(figsize=(8,5))
sns.violinplot(data=df,x='time',y='total_bill',hue='sex',split=True,palette='coolwarm')
plt.title('Spend Distribution Shape by Meal Time and Gender (Customer Analytics)')
plt.xlabel('Meal time')
plt.ylabel('Spend ($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```



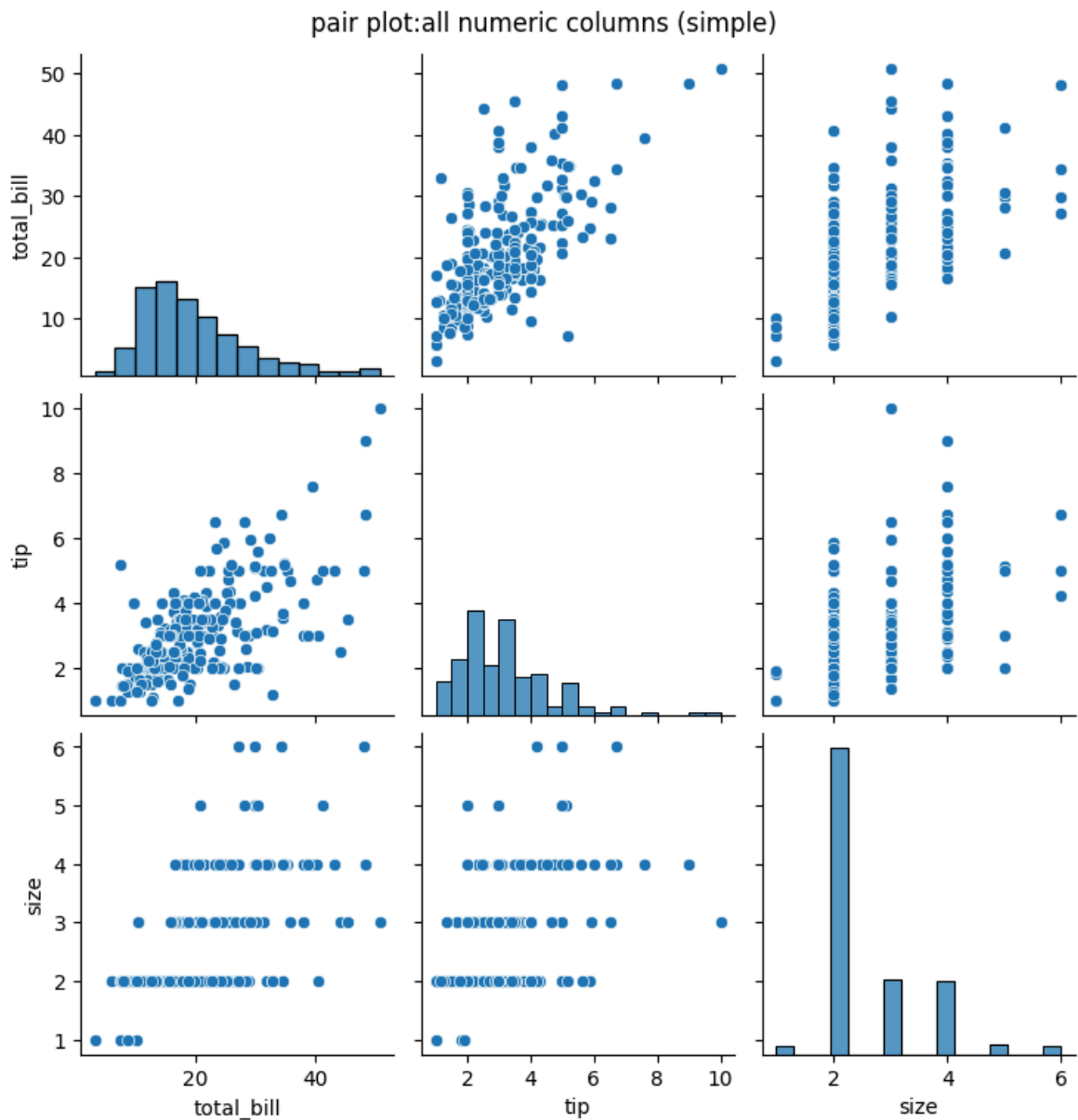
## 12. Pair Plot-sns.pairplot()

```
In [51]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [52]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [53]: sns.pairplot(df)
plt.suptitle('pair plot:all numeric columns (simple)',y=1.02)
plt.show()
```



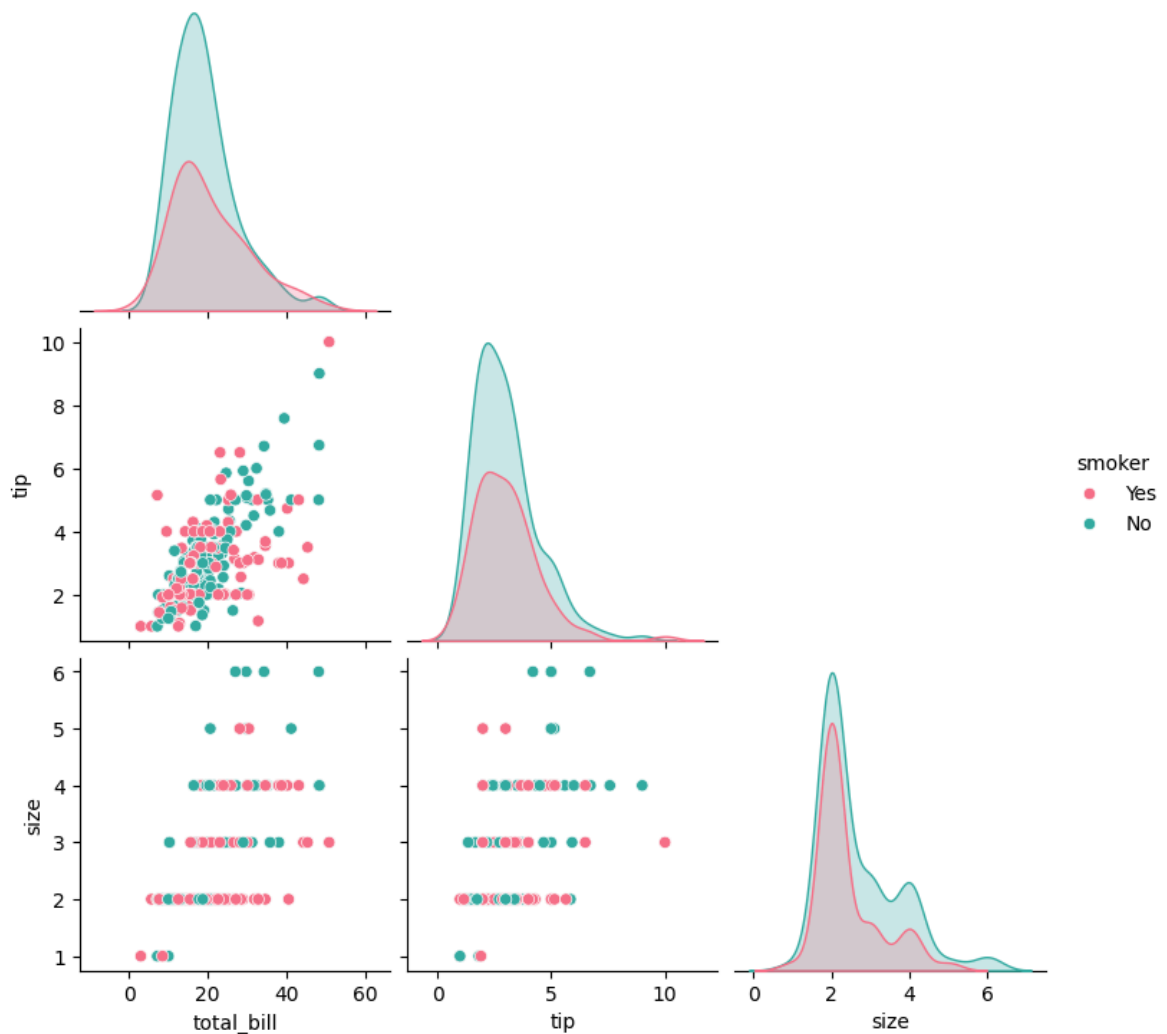
```
In [54]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [55]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [56]: sns.pairplot(df,hue='smoker',corner=True,diag_kind='kde',palette='husl')
plt.suptitle('pair plot:corner view with KDE Diagonal,by Smoker Status',y=1.02)
plt.show()
```

pair plot:corner view with KDE Diagonal,by Smoker Status

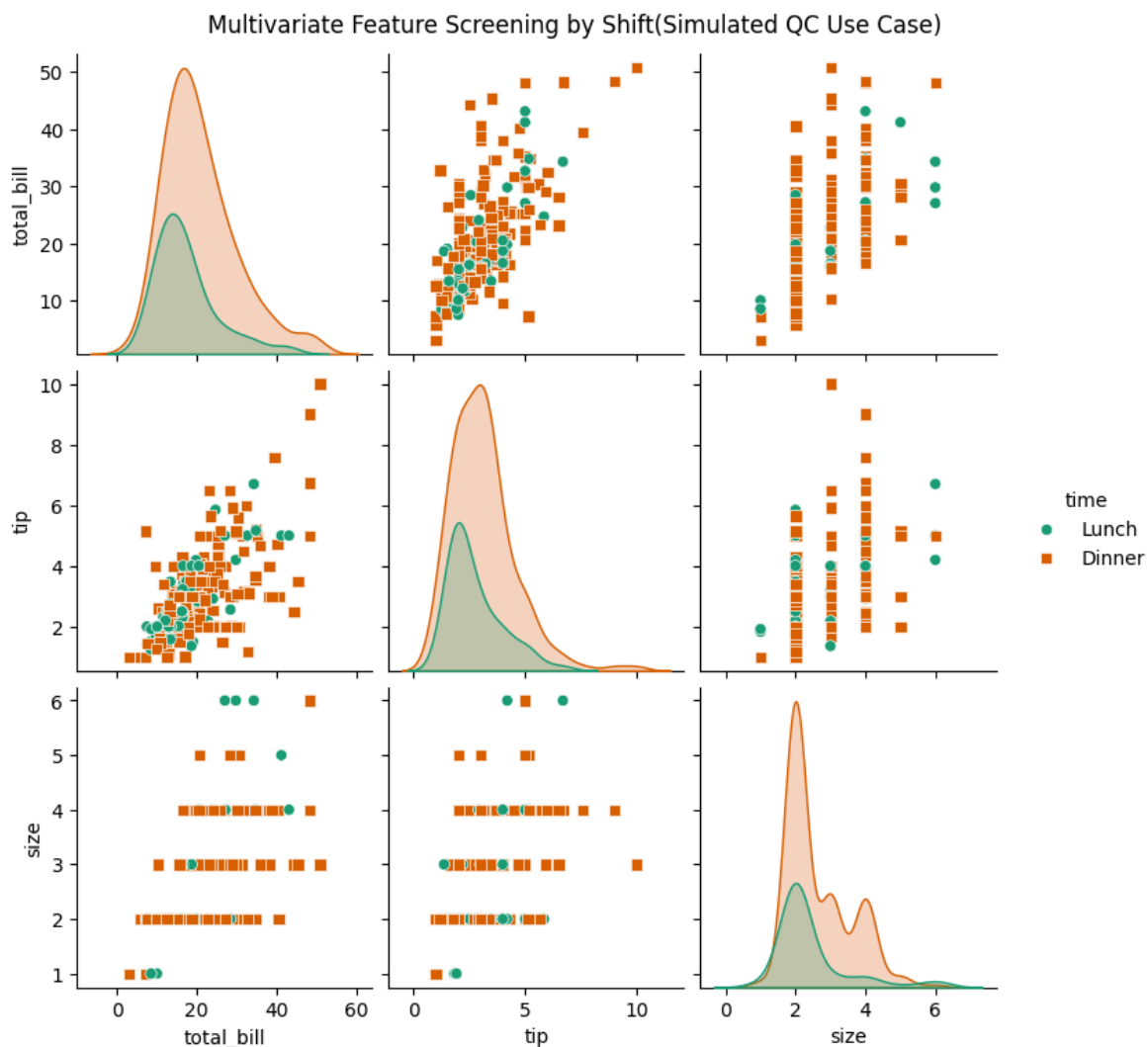


```
In [57]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [58]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [59]: sns.pairplot(df,vars=['total_bill','tip','size'],hue='time',markers=['o','s'],pa
plt.suptitle('Multivariate Feature Screening by Shift(Simulated QC Use Case)',y=
plt.show()
```



13.Heatmap-sns.heatmap()

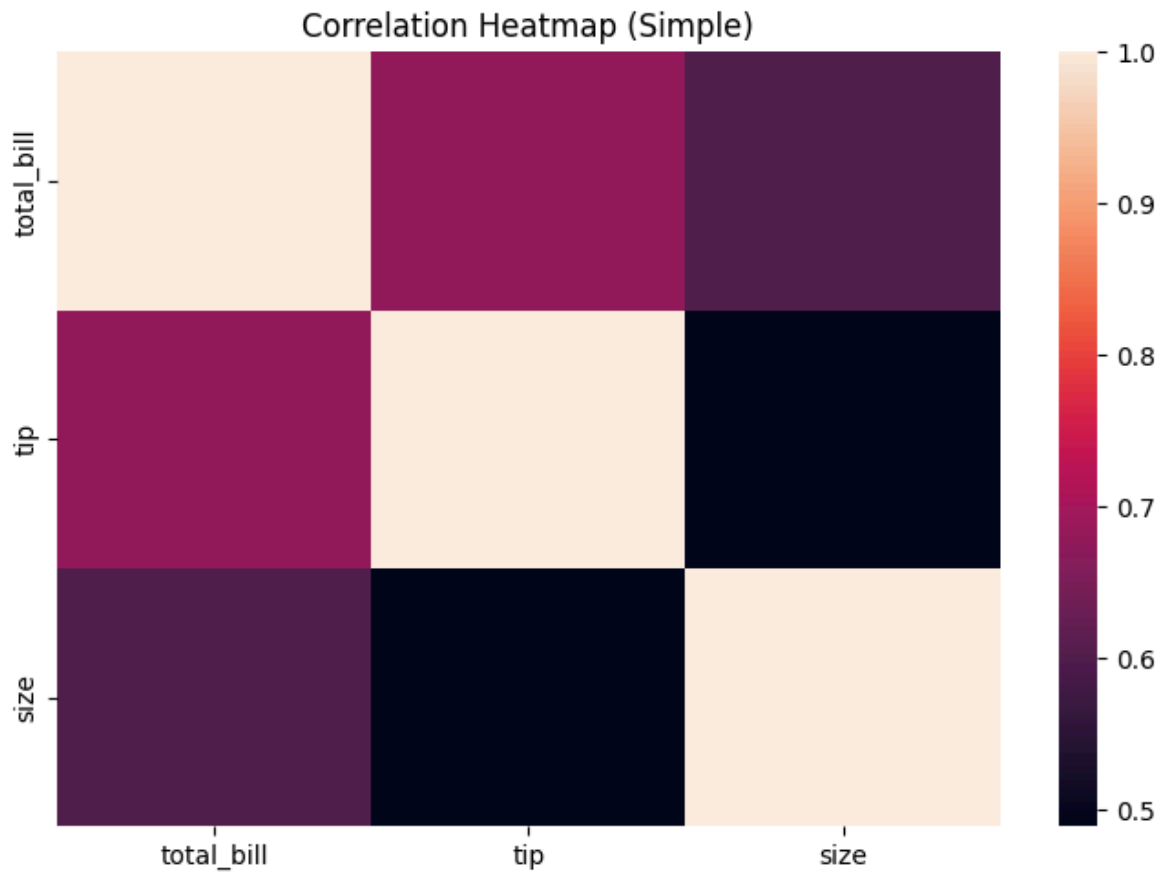
```
In [60]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [61]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [62]: corr=df.corr(numeric_only=True)

plt.figure(figsize=(7,5))
sns.heatmap(corr)
plt.title('Correlation Heatmap (Simple)')
plt.tight_layout()
plt.show()
```

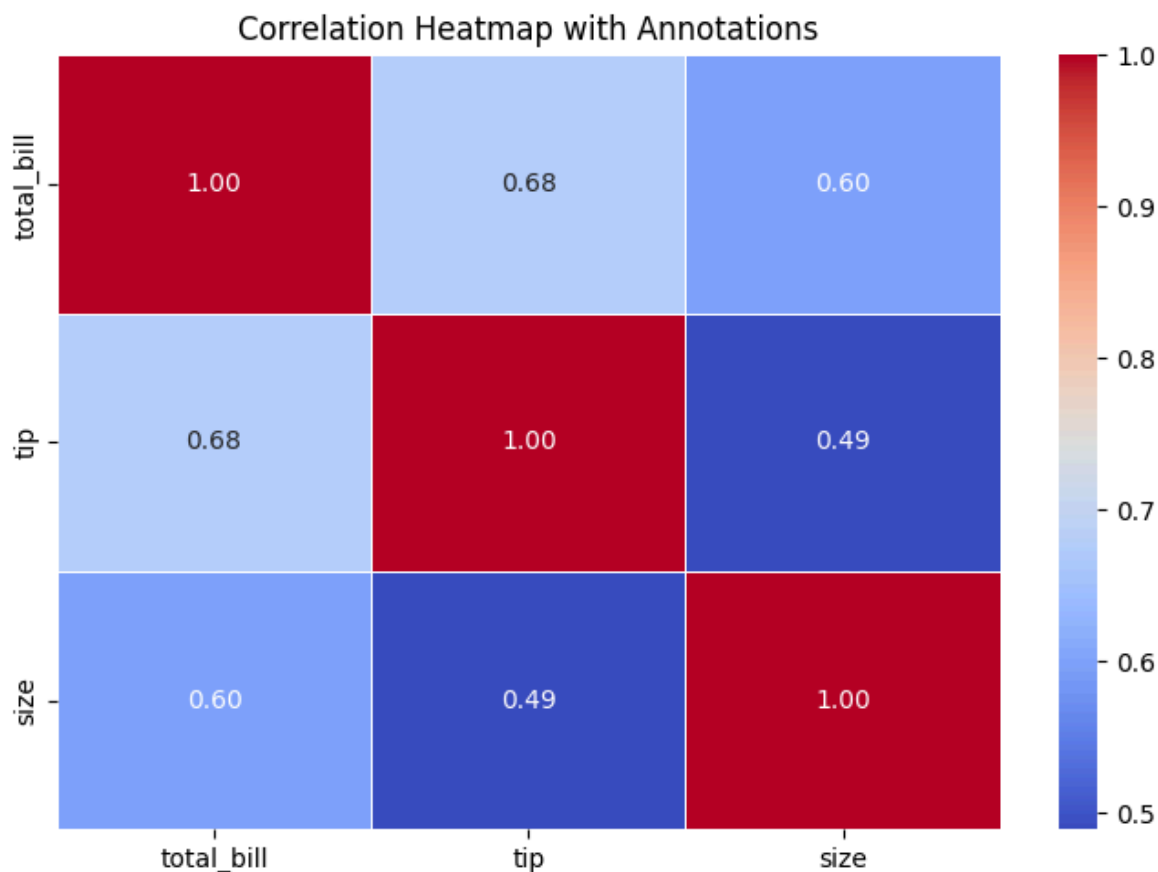


```
In [63]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [64]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [65]: plt.figure(figsize=(7,5))
sns.heatmap(corr,annot=True,cmap='coolwarm',fmt='.2f',linewidths=0.5,cbar=True)
plt.title('Correlation Heatmap with Annotations')
plt.tight_layout()
plt.show()
```



```
In [66]: import matplotlib.pyplot as plt
import seaborn as sns
```

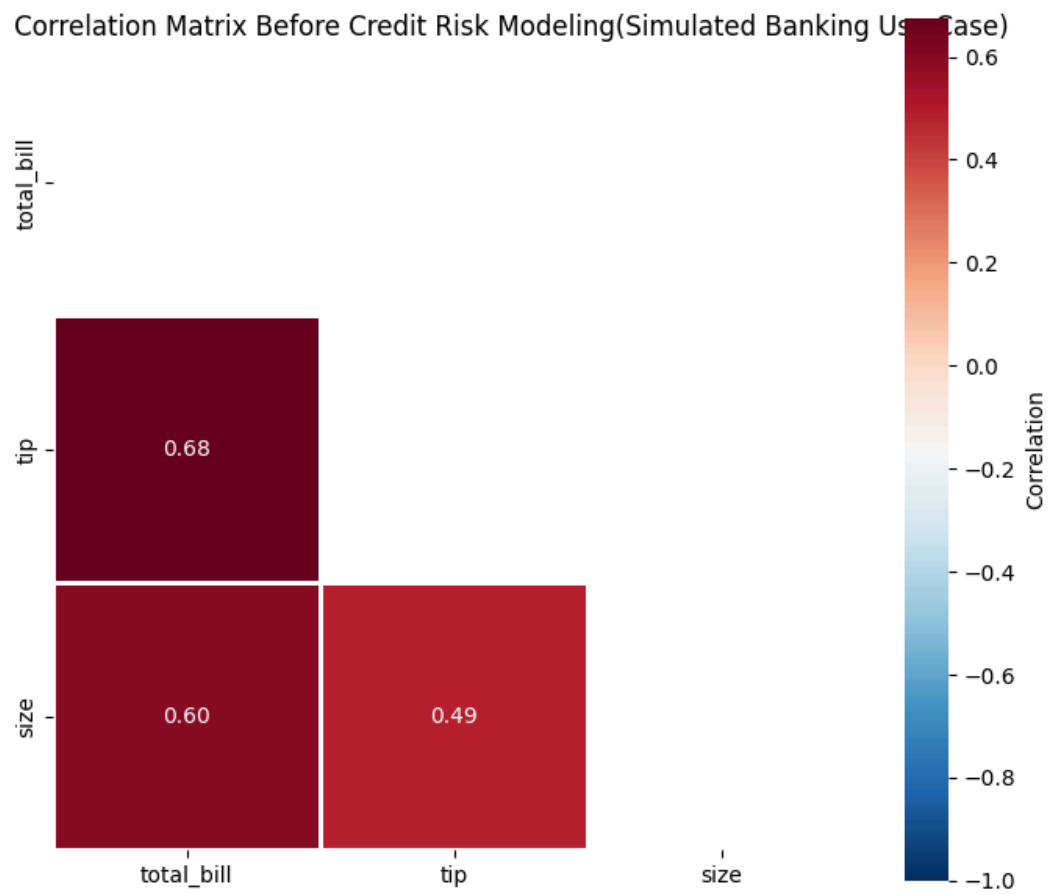
```
In [67]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")
```

```
In [68]: corr=df.corr(numeric_only=True)
mask=np.triu(np.ones_like(corr,dtype=bool))

plt.figure(figsize=(7,6))
sns.heatmap(corr,annot=True,cmap='RdBu_r',fmt='.2f',linewidths=0.8,mask=mask,vmi
plt.title('Feature Correlation Matrix Before Credit Risk Modeling(Simulated Ban
plt.tight_layout()
plt.show()
```

Feature Correlation Matrix Before Credit Risk Modeling(Simulated Banking Use Case)



In [ ]: